

Zero-Shot Learning for Computer Vision Applications

Sandipan Sarma

Indian Institute of Technology Guwahati

Guwahati, Assam, India

sandipan.sarma@iitg.ac.in

ABSTRACT

Human beings possess the remarkable ability to recognize unseen concepts by integrating their visual perception of known concepts with some high-level descriptions. However, the best-performing deep learning frameworks today are supervised learners that struggle to recognize concepts without training on their labeled visual samples. Zero-shot learning (ZSL) has recently emerged as a solution that mimics humans and leverages multimodal information to transfer knowledge from seen to unseen concepts. This study aims to emphasize the practicality of ZSL, unlocking its potential across four different applications in computer vision, namely – object recognition, object detection, action recognition, and human-object interaction detection. Several task-specific challenges are identified and addressed in the presented research hypotheses. Zero-shot frameworks are proposed to attain state-of-the-art performance, elucidating some future research directions as well.

CCS CONCEPTS

• **Computing methodologies** → **Transfer learning**; *Object recognition; Object detection; Activity recognition and understanding; Scene understanding.*

KEYWORDS

Zero-shot learning, seed construction, visual-semantic mining, GAN, triplet loss, transformer, CLIP, object recognition, object detection, action recognition, human-object interaction detection

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1 INTRODUCTION

Motivation. A significant proportion of the impressive performance of deep learning models today can be attributed to the availability of enormous amounts of annotated data used for supervised training. However, this is also a limitation in practical scenarios, where it is impossible to collect annotated data for every *concept* (an object, action, or interaction) in the world. Zero-Shot Learning (ZSL) helps consolidate computer vision models for the real-world

better, where the visual knowledge gained from a collection of a few classes of concepts can be combined with attribute information of similar classes, even if novel, to recognize or detect them when deployed in the real world. Hence, the dependency on labeled data can be reduced while retaining the performance gains of deep learning methods to a certain degree, turning ZSL into one of the most sought-after strategies for the future.

Being a learning paradigm, ZSL can be applied to several computer vision applications, the most fundamental of them being object recognition, i.e., predicting a single class label given an image. In **zero-shot object recognition (ZSR)**, semantic descriptions (attributes) are available as an auxiliary source of information, but visual samples are available only for *seen* object classes. Despite methods improving the alignment between visual-semantic spaces [8] and learning data distributions conditioned on object semantics to generate visual data for unseen objects [27], these models train on seen classes manually predetermined by [45]. But these splits are not designed for optimal zero-shot performance [20]. It is necessary to ensure that the diversity and rarity of object domains are well-captured in the designated seen classes for training ZSR models.

The ability of humans to analyze motion patterns and perceive an action subsequently follows their understanding of object recognition. Naturally, human action recognition in videos has gained immense popularity, especially today when video accounts for about 82% of the global IP traffic [5]. Despite this huge availability of video data, their unstructured nature and time-consuming annotation process call for **zero-shot action recognition (ZSAR)** to help recognize the ever-increasing number of actions. Beyond the frameworks learning from the human pose [14], improving visual [16] and semantic [9] representations, and distilling knowledge from graphs [11], pretrained image-based visual-language models successfully adapted to ZSAR are the best performers currently. Temporal modeling [18] and video-specific prompting mechanisms [28] via transformers have massively boosted classification accuracy. However, they are limited when it comes to capturing the local spatial context that can be mined from objects and action environments. Moreover, widely-used semantic embeddings like Word2Vec [25] are considerably noisy, and modeling action relationships with a richer representation is desirable. On the other hand, actions are also human-centric from a zero-shot perspective, where relative temporal motions between human body parts can be the point of knowledge transfer from the seen to unseen actions.

Zero-shot object detection (ZSD) is another challenging area since the localization of objects is also important in addition to correct object classification. Similar to ZSR, the most common approach for ZSD is improving visual-semantic alignment using pretrained models like Faster-RCNN [34] as backbone object detectors. However, it suffers from the hubness problem [31], where a target

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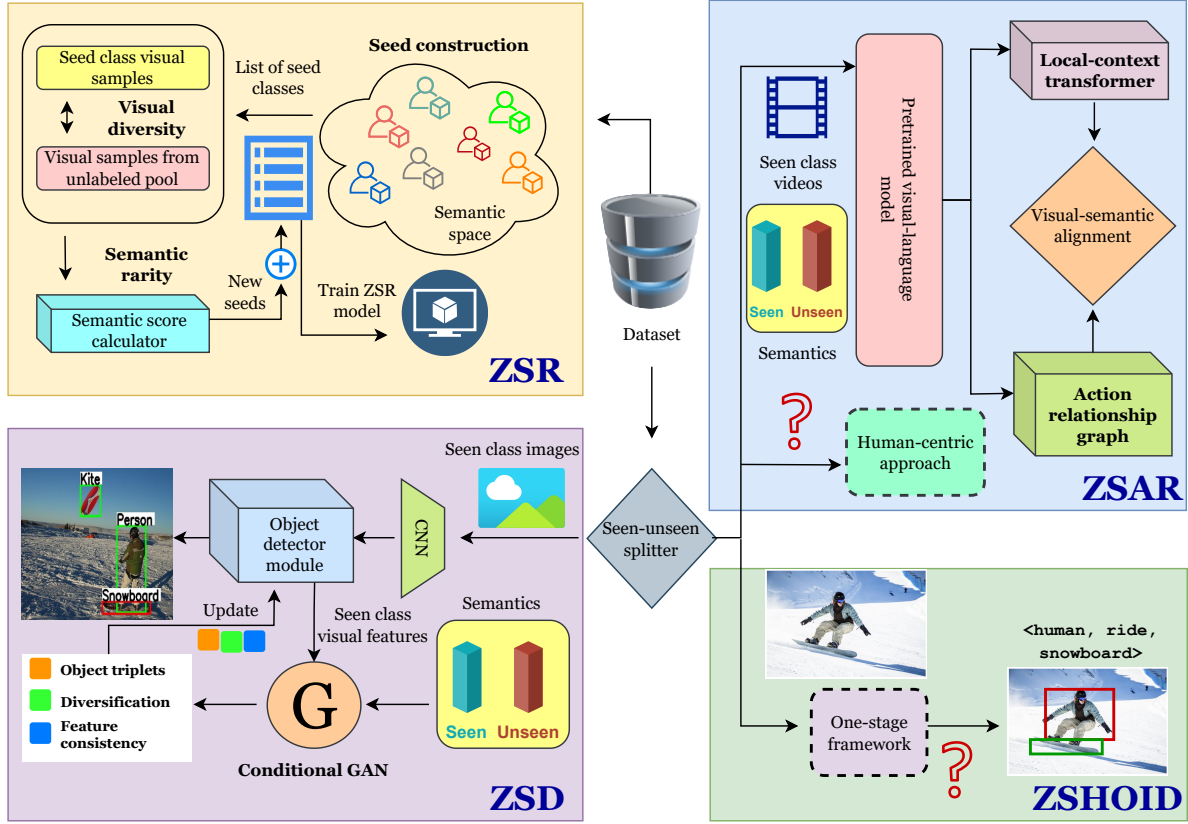


Figure 1: The research problems in four different zero-shot applications and their frameworks envisioned in this doctoral study. The dotted blocks with question marks are the tasks in progress or planned for the future.

class becomes the nearest neighbor for most of the region proposals during testing. Generative frameworks like [15] resolve this problem by generating visual features for unseen classes, but they are conditioned on class semantics. Consequently, the synthesized object features for semantically-similar object classes like *car* and *bus* often confuse the zero-shot object detectors.

In scene understanding, **zero-shot human-object interaction detection (ZSHOID)** is an emerging and challenging task where rather than simply detecting objects, we need to detect a human, the object(s) he/she interacts with and predict the interaction label as well. Being compositional in nature, this task suffers from the long-tail distribution problem [46], which means that barring some common interactions, most human-object compositions like “*a person kissing a tiger*” have very few training examples. By definition, ZSL seems like a suitable answer to this problem. The intuitive idea of using the human pose to guide HOI prediction [12] can misfire due to the polysemy of action verbs, where the meaning of a verb and the related human pose can change with context. Although some methods generate object features for unseen HOI composition [17], it is difficult to generate verb representations.

Problem description. The goal of this doctoral research is to establish the benefits that we can reap from zero-shot approaches by identifying some crucial research gaps in the four aforementioned

vision applications (Figure 1). While the task complexity increases from object recognition to HOI detection, the size of available annotated data also decreases. For ZSR, an intelligent class selector is designed that can select the suitable classes automatically from a collection of objects to train any (existing or upcoming) attribute-based ZSR model and boost its performance. A generative framework for ZSD is designed that maintains visual-semantic cyclic consistency and mitigates confusion between semantically-close object features via a triplet loss. A two-stage method is explored to aggregate better spatiotemporal embeddings and integrate them with enriched semantic representations of actions in ZSAR.

2 RELATED WORK

Zero-shot object recognition (ZSR). Lampert et al. [19] were one of the first to come up with the idea of formalizing zero-shot learning. Their approach of learning probabilistic attribute classifiers was quite popular initially for ZSR [20] until it was overshadowed by compatibility learning functions (CLFs) [1] that measure the compatibility between visual and semantic spaces via trainable weights. With the increasing popularity of generative frameworks, they were adapted in ZSR [27] to synthesize high-quality unseen class visual features, eliminating the hubness problem [31] faced by CLFs and promoting generalizability. More recently, knowledge

graphs [47] and transformers [7] have also been used to improve ZSR accuracy. However, all of these approaches use predetermined seen-unseen splits [45] for ZSR model evaluation. Wang et al. [40] automatically select the suitable classes for training a ZSR model, but their algorithm is limited to GCN-based frameworks [43].

Zero-shot object detection (ZSD). Since the formalization of ZSD in 2018 [33], the usual approach has been to train an off-the-shelf object detector like Faster-RCNN [34] on seen class data to extract object features and then learn visual-semantic projection functions. Various task-specific loss functions have been designed for ZSD. For instance, the max-margin loss [4] handles the problem of background-unseen confusion, region scoring [10] ensures compatibility between object regions and class semantics, and polarity loss [32] enforces a proper penalization of a training sample considering the background-foreground imbalance. As in ZSR, generative frameworks [15] have proven to be one of the best frameworks, but producing features of unseen class objects that are discriminative enough still remains a problem.

Zero-shot action recognition (ZSAR). The earliest approaches focus on improving visual-semantic associations [22] by learning better semantic embeddings via textual descriptions [39], leveraging synonyms [2], or constructing action descriptions [9]. On the other hand, attempts have been made to improve visual representations via generative networks [16]. In addition to visual-semantic knowledge, the idea of exploiting additional human pose information and modeling it using graph networks has proven beneficial. However, even the best transformer-based networks [18, 28] today ignore local contextual cues from objects and environments.

Zero-shot human-object interaction detection (ZSHOID). Shen et al. [37] were the first to demonstrate HOI detection performance in both fully-supervised and zero-shot settings via a novel verb-object disentangling network, and several other works followed a similar pipeline [3, 42]. Usually, these are two-stage networks that detect humans and objects first and generate human-object matched pairs, which are then fed to an interaction classifier. Despite some task-specific novelties like fabricating object features [17] and consistency graphs [23], the dependency of the interactive classifier on the outputs from the previous stage restricts its potential. Recently, one-stage methods built on the top of vision transformers [41] and DETR [44] show significant improvements.

3 RESEARCH QUESTIONS

This section introduces the four research questions being investigated and their potential impact on the target vision applications.

- (1) **Do diversity and rarity in an object domain affect a ZSR model's ability to recognize unseen objects in the wild?** Collecting annotated data for every object in the world is not feasible. Hence, for research purposes, data can be collected for only a fixed (say N) number of available categories. However, ZSL researchers require a disjoint *seen-unseen* split of these N classes, which are currently predetermined [45] for a few datasets (we call them **Existing Splits (ES)**). Most existing ZSR methods learn from these seen classes only. However, there are two problems with ES. Firstly, the manual splits are not necessarily the best ones for ZSR [20], and their seen classes (used for training) may not capture the

visual diversity and semantic rarity of an object domain (say “birds”). And secondly, ES is not defined for all datasets. To experiment on a new dataset $\mathcal{D}_{new}$, one has to either evaluate their methods on k different random seen-unseen splits and report the average accuracy [6] or manually re-define a set of benchmark seen-unseen splits of $\mathcal{D}_{new}$ meticulously like [45], which is impractical. Therefore, we look to intelligently yield a set of classes from a given dataset for ZSR training. Since improving the accuracy by even 1-2% is considered a good stride towards improving ZSR as a whole, we explore whether or not by having a good initial idea of the diversity and rarity in an object domain, a ZSR model can show better transferability from seen to unseen classes.

- (2) **How to mitigate the semantic confusion between similar objects for ZSD?** Although they exhibit arguably the best generalizability in ZSD literature, existing generative frameworks like conditional GANs [13] only learn how to synthesize image features based on class semantics [15], disregarding the degree of dissimilarity between object classes while training. When visual features of an unseen class (e.g., *bicycle*) are generated by such a GAN, it might be confused with object features of a semantically similar seen class (like *motorcycle*), resulting in semantic confusion and wrongly detecting the object. Moreover, if the synthesized visual features of an object do not correspond well to its semantic representation, such unconstrained GAN training might produce inconsistent visual features for the unseen classes. This would ultimately impact the seen-unseen understanding of a detector like Faster-RCNN, leading to misclassifications or even misinterpreting unseen objects as “background” at test time. Therefore, the goal is to handle these two aspects while training a GAN.
- (3) **How to extract rich contextual information from the human body, objects, and environments for ZSAR?** When distinguishing two actions, the object of interaction can play an important role even if the spatiotemporal motion trajectories of the body parts are relatively similar (e.g., throwing a *hammer* versus a *discus*). Even for differentiating actions independent of any object of interest, an additional semantic extracted from video frames can be useful (e.g., understanding “the presence of water” can differentiate *breast-stroke* and *horse riding*). On the other hand, humans also tend to interact similarly with functionally-similar objects (*punching bags* and *speed bags*) and environments (e.g., *aquatic surroundings*). Therefore, contextual cues from objects and environments play a dual role – fuelling both distinguishability and functional similarity. Additionally, modeling the relative motion between human body parts can facilitate contextual knowledge transfer from a seen (e.g., *playing golf*) to an unseen action (e.g., *baseball shot*).
- (4) **How to generate proper verb representations and attain a better notion of “interactiveness” for ZSHOID?** HOIs are compositional in nature. For instance, with the verb “riding”, several HOIs can be composed like *riding horse*, *riding bike*, *riding snowboard*, etc. However, unlike other applications, generating visual representations is not straightforward here as verbs are not well-defined like objects and

can have dynamic meanings as per the interaction context (the polysemy problem). Hence, there is a need to generate data-driven verb representations dynamically that can supplement the interaction features obtained from human-object pairs by a pretrained object detector.

4 PROPOSED APPROACHES, EXPERIMENTS, AND RESULTS

We first propose a framework called **DiRaC-I** [36] inspired by active learning that captures the diversity and rarity of the object domain within a set of seen classes, using which any attribute-based ZSR model can be trained to have a comprehensive idea of the domain. The only work closest to ours is GAZL [40], but its main limitations are: (1) visual samples from unseen classes are also used during the training phase for validating the model performance, which is impractical for real-life class selection; (2) external sources of information [26] are used to establish class relationships; and (3) it is not a method *dedicated* to finding the most suitable training classes. In our two-stage framework, the first stage is seed-set construction, where N classes of a given dataset are clustered based on semantic similarity. A single representative is picked from each cluster to ensure diversity while jointly prioritizing semantic rarity, forming a seed set. The seed set initializes the second stage – visual-semantic mining – where we seek to maximize the diversity between visual samples of the seed and non-seed classes by estimating the distribution of related classes via activation vectors. This selects a few candidate non-seed classes and a semantic score is computed for each of them. A few classes having the highest semantic scores are added to the seed set. This process continues till we get a fixed S number of classes ($S \ll N$) as the *seen classes* for the given dataset. Experiments on two benchmark datasets and across more than ten ZSR models demonstrate seen classes picked by DiRaC-I help ZSR models to achieve significant classification accuracy improvements as compared to ES – up to 8% and 5% for CUB [38] and SUN [29], respectively. Additionally, while recognizing classes exhibiting rare attributes we get performance boosts of up to 10% and 7% for CUB and SUN respectively. Hence, acknowledging rarity yields better performance for novel and rare classes, establishing our framework’s convenience in the wild.

To answer research question 2, we devise a conditional GAN to synthesize unseen class features constrained by a new triplet loss where the triplets are strictly visual feature-based. We believe that the quality of visual features generated on the basis of class semantics heavily impacts the classification ability of a backbone object detector at a later stage. Therefore, accounting for inter-class dissimilarities explicitly in the visual space can address semantic confusion during the feature generation phase itself. Moreover, unlike the traditional fixed margin of separation between positive and negative triplets, we use a flexible margin acquired as a pre-computed value from a rescaled Mahalanobis distance matrix with respect to class semantics. This encourages the GAN to acknowledge the degree of dissimilarity between two classes and reflect them in the generated features by accounting for the first and second-order statistics of the class semantics. Additionally, a cyclic consistency module enforces the generated features to reconstruct the corresponding class semantics. Regularized by a mode-seeking

loss [24], the final generated features are diverse and robust to semantic confusions and visual-semantic inconsistencies. Our method achieves a relative mAP gain of 8% and a relative Recall@100 gain of 6% over the existing state-of-the-art. In the generalized ZSD setting on MS-COCO [21], we get a relative mAP gain of 10.5%. Qualitative analysis and ablation studies in this **published work** [35] illustrate the effect of loss components and a good separability between generated features of semantically-similar classes.

Motivated by the dual role of object-centric and environmental context for ZSAR, we explore research question 3 from both the visual and semantic perspectives. Additionally, we use a pretrained CLIP [30] as a source of prior knowledge. From the visual perspective, we construct a temporal transformer $\mathcal{T}$ that takes the frame-level visual embeddings extracted using CLIP as input. Within $\mathcal{T}$, we integrate a context aggregating module that looks for local contextual cues from multiple scales and produces a more informative visual embedding for a video sample. From the semantic perspective, how well we capture class relationships plays a vital role in how strong the semantic bridge between seen and unseen actions will be. To this end, we propose to model action relationships via a Graph Attention Network that learns proper statistical strengths for different neighbors of an action node and produces semantic embeddings. A strong alignment between the obtained visual-semantic embeddings helps us beat the state-of-the-art.

5 REPRODUCIBILITY

To ensure that the outcomes of this doctoral study are accessible to the ZSL community around the world, the Python-based implementations of the undertaken research are open-sourced on GitHub. PyTorch, a widely-popular deep learning framework, is used for all the experiments for better outreach. The codes are well-commented and step-by-step instructions are given for reproducibility. The pretrained models, data splits, and log files are also uploaded.

6 CONCLUSION AND FUTURE DIRECTIONS

In this doctoral study, four research questions are addressed, and a few novel ideas are proposed. An active learning-inspired intelligent framework has been developed to acknowledge the visual diversity and semantic rarity of an object domain to suggest which classes are the most suitable and informative to be used for training ZSR models. Then, the outputs of a conditional GAN are constrained via a triplet loss with a semantically-controlled margin and a cyclic consistency loss to maintain visual-semantic consistency in the synthesized features, which infuse unseen class knowledge in object detectors for ZSD. For research question 3, we find that extracting local visual context via a novel temporal transformer and pairing it with action relationships provide a better visual-semantic alignment for ZSAR. We are currently working on a human-centric ZSAR approach, where we construct novel geometric representations of human body parts and use them as complementary information for strengthening our visual-semantic embeddings. In the future, the plan is to investigate how to generate robust verb representations from a given image for ZSHOID and improve the notion of “inter-activeness”, with the aim of obtaining state-of-the-art results in the fully-supervised setting as well.

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